

FORM TP 02025092



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CARIBBEAN EXAMINATIONS COUNCIL

**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 02 – General Proficiency

2 hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of FOUR questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
5. **If you use the extra page(s), you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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01229020/MJ/CSEC 2025



0 1 2 2 9 0 2 0 0 3

Answer ALL questions.

1. (a) Peripheral devices are those which are usually added to a computer system, connected to and controlled by the CPU.

- (i) Circle TWO peripheral devices in the list of devices below.

Motherboard

Microphone

Monitor

Keyboard

Compact disc

Mouse

(2 marks)

- (ii) For the TWO peripheral devices circled in (a) (i), state how EACH may be used by an individual in the music industry.

Microphone and Keyboard

(2 marks)

- (b) Printers are output devices used to create hard copy documents.

- (i) Name ONE of the two categories of printers.

Impact

or

Non Impact

(1 mark)

- (ii) Outline ONE difference between a dot matrix printer and a thermal printer.

A dot matrix printer uses a print head that strikes an inked ribbon against paper to create prints

while a thermal printer uses heat to produce an image

(2 marks)



- (c) A musician is in the process of sourcing a printer. Provide THREE reasons why a laser printer should be chosen over an inkjet printer.

Cost per page: Laser printers are cheaper since they use lasers while inkjet uses ink cartridges to print.

Speed: Laser printers are faster at printing out large volumes of music sheets

Durability: Laser prints are often more resistant to smudging and water damages

(3 marks)

- (d) Consider the following database with two tables named **FARMERS** and **FARM**.

FARMERS

ID No.	FirstName	LastName	Gender	Contact Number	Region
1	Jack	Smith	M	675-8742	1
2	Leslie	Jacobs	F	782-6734	2
3	Bruce	Jordan	M	892-6764	1
4	Karen	Smith	F	323-8792	7
5	Linda	Daniels	F	239-0552	2
6	Jerome	Levy	M	982-1291	5
7	Carlington	Nichols	M	767-2231	5
8	Handel	Brown	M	773-2231	4

FARM

FarmID	Crop	Cost Per Produce	FarmerID
002	Banana	\$10.00	1
192	Orange	\$8.25	1
778	Irish potato	\$30.00	6
252	Lychee	\$25.00	2
123	Sweet pepper	\$15.00	5

- (i) In the database table FARM, name the field by which EACH record can be uniquely identified.

FarmID

(1 mark)

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



0 1 2 2 9 0 2 0 0 6

- (ii) State the fields that must be linked in order to create a relationship between the two tables.

ID No. linked with the FarmerID

(2 marks)

- (iii) Complete the Design View extract below to show a simple query that will list the Farmer's full name, Region, Crop and FarmID for FarmerID number 1.

Field						
Table						
Sort						
Show	☐	☐	☐	☐	☐	☐
Criteria	1					
Or						

(4 marks)

- (iv) Complete the Datasheet View extract below by displaying the result of the query created in (d) (iii).

FirstName	LastName	Region	Crop	
Jack	Smith	1	Banana	
Jack	Smith	1	Orange	

(3 marks)

Total 20 marks

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



2. (a) Define EACH of the following terms.

- **Verification**

The process of checking that data has been copied accurately from one medium to another.

- **Validation**

An automatic computer check to ensure that the data entered is sensible and readable.

(2 marks)

(b) State ONE example of a typical data entry error.

Transposition error

(1 mark)

(c) For EACH validation check in the following table, state ONE example of its use.

Validation Check	Example of Use
Range check	Ensuring a student's test score is between 0 to 100
Data type check	Ensuring a Quantity field is only contains correct numbers
Presence check	Ensuring the middle name field is not left blank on a form

(3 marks)

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



0 1 2 2 9 0 2 0 0 8

- (d) (i) Describe how a double data entry check can be applied on a student's application form with respect to confirming the student's age.

A password is entered twice. Then the system is going to compare them both
if they do not match, then the system would alert the user to type it again

(1 mark)

- (ii) For EACH of the following scenarios, state the type of validation check that should be used.

- (a) The age entered must correspond with the age calculated from the date of birth.

Consistency Check

- (b) The national insurance number must have three alphanumeric characters followed by six numeric characters, for example, XXX-999999.

Format Check

- (c) A password must have a specific number of characters.

Length Check

(3 marks)

- (e) Jane is writing an essay using a word processing software. Describe the steps that Jane should take to determine the number of words in her essay.

Highlight the text and check the status bar at the bottom of the window, or go to the review tab and click word count

(2 marks)

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



0 1 2 2 9 0 2 0 0 9

- ## Hyper-text markdown language

HTTP

Mail merge

Web Technology Concept	Description
HTML	A language used to create documents viewed on the World Wide Web.
Hyperlink	A clickable word or image that opens a file or application.
Mail Merge	A document is created as a template and is used to create multiple personalized documents for distribution.
Web browser	A program used to view information on the World Wide Web.
Webpage	A single document on the internet.

(g) The following URL is used by students to gain access to their examination results.

Describe EACH of the THREE components of the URL.

*www.cxc.org is the unique address of the website on the internet (domain).....

(3 marks)

Total 20 marks

GO ON TO THE NEXT PAGE



3. Manufacturer X is in the process of integrating technology into its day-to-day operations and has hired different consultants to assist in the selection of the correct hardware and software. The company has also hired programmers who will be developing an application specifically for its use.

- (a) State ONE difference between hardware and software, and provide an example of EACH.

Hardware refers to the physical components of the computer you can touch

Software refers to the set of instructions or programs that tell the hardware what to do

Hardware: Hard drive

Software: Windows OS / Operating System

(3 marks)

- (b) (i) Identify the category of the software being created for Manufacturer X.

Custom-written or bespoke software

(1 mark)

- (ii) A member of the accounts department is having trouble printing a document from the current payroll application. She printed the document but only saw lines on the paper.

State the likely cause of the problem.

The wrong printer driver is likely installed

or

The cables are loose

(1 mark)

- (c) The application would be designed for the interaction of users via a graphical user interface.

State TWO advantages of a graphical user interface over a command-line user interface.

Ease of use, since GUIs are visual icons and pointers

Multitasking because GUIs allows users to easily switch between multiple open windows and applications

(2 marks)

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



0 1 2 2 9 0 2 0 1 1

- (d) A 2 terabyte solid state drive will be installed in the computer of the lab technician at Manufacturer X. Determine the number of 5 gigabyte files that could fit on the drive. **Show all working.**

Total Space: 2 TB = 2000 GB

File Size: 5 GB

Calculation: $2000 / 5 = 400$ Files

(3 marks)

- (e) Having met with the chief technical officer for Manufacturer X, the programmers/designers will finalize the problem-solving process.

- (i) Place EACH of the following problem-solving steps in the correct order.

- Define the problem.
- Develop the algorithm.
- Propose and evaluate solutions.
- Test and validate the solution.
- Determine the most efficient solution.

Define the problem

Propose and evaluate solutions

Determine the most efficient solution

Develop the algorithm

Test and validate the solution

(3 marks)

- (ii) Complete the paragraph below by inserting the most appropriate terms in the blank spaces.

The values that are changeable during the execution of the program are known as

Variable

and the values set by the programmer that will never change

during the execution of the program are known as Constants

(2 marks)

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



0 1 2 2 9 0 2 0 1 2

- (f) Complete the following algorithm snippet to show the payroll process for Manufacturer X.

Rate $\leftarrow$ 5

ExpectedHours $\leftarrow$ 40

OvertimeRate $\leftarrow$ 6

Read hoursWorked

IF hoursWorked > ExpectedHours **THEN**

OvertimeHours $\leftarrow$ hoursWorked - Expected Hours

Pay $\leftarrow$ (OvertimeHours * OvertimeRate) + (Expected Hours * Rate)

ELSE

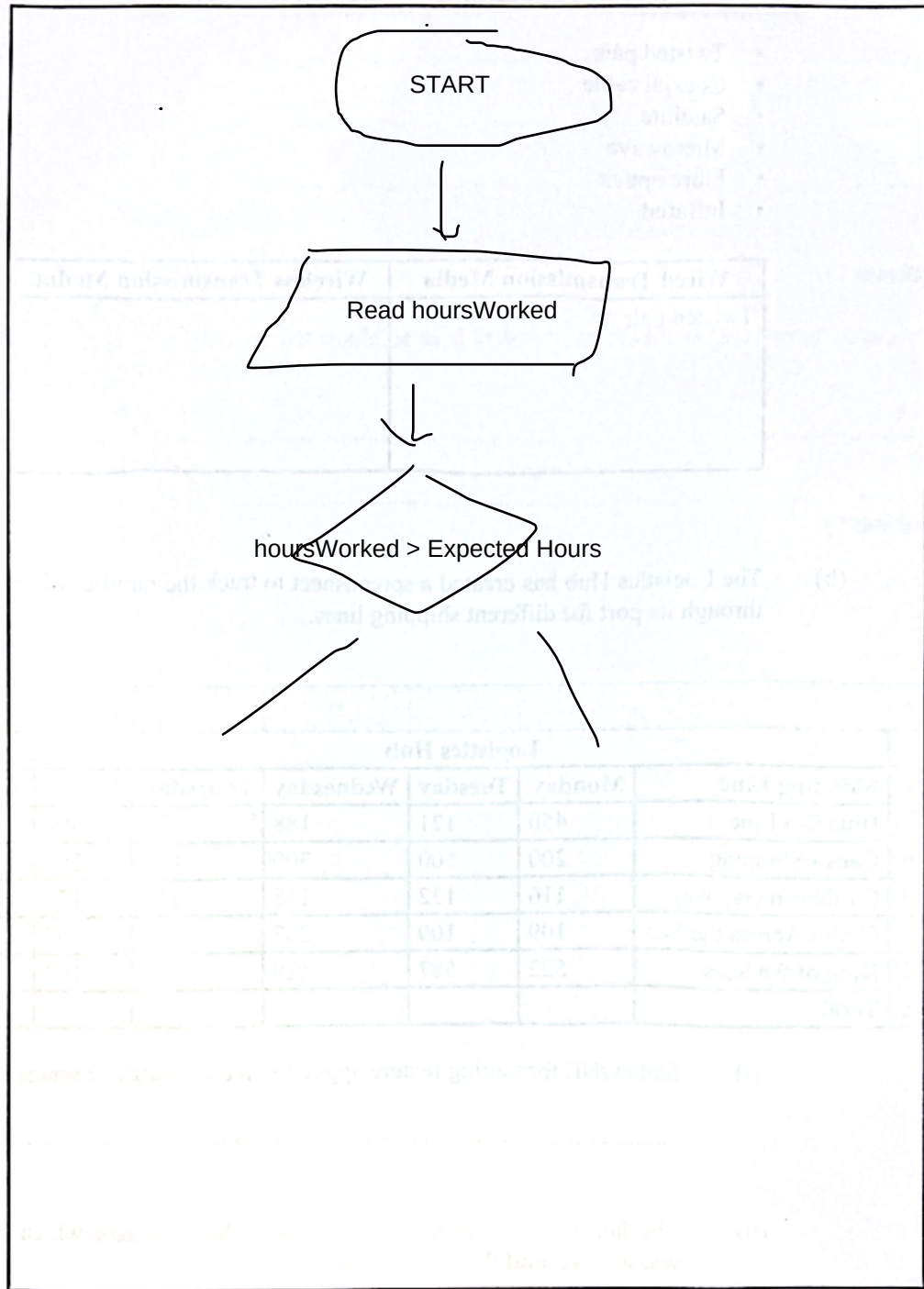
Pay $\leftarrow$ hoursWorked * Rate

ENDIF

(3 marks)



- (g) Draw a flowchart to represent the completed algorithm snippet in (f) on page 14.



(7 marks)

Total 25 marks

01229020/MJ/CSEC 2025

GO ON TO THE NEXT PAGE



4. (a) Complete the table below by placing EACH of the following transmission media devices into the appropriate category. (An example is given.)

- Twisted pair
- Coaxial cable
- Satellite
- Microwave
- Fibre optics
- Infrared

Wired Transmission Media	Wireless Transmission Media
Twisted pair Coaxial cable Fibre optics	Satellite Microwave and Infrared

(5 marks)

- (b) The Logistics Hub has created a spreadsheet to track the number of containers that pass through its port for different shipping lines.

	A	B	C	D	E	F	G	H
1	Logistics Hub							
2	Shipping Line	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Total
3	Blue Sea Line	450	121	188	290	501	420	
4	Canva Shipping	200	500	300	125	203	112	
5	Caribbean Gateway	116	132	165	111	178	101	
6	Grains Across the Sea	109	100	207	186	232	200	
7	King of the Seas	522	587	429	139	192	100	
8	Total							

- (i) State ONE formatting feature applied in the spreadsheet image above.

Bold text (used in headers)

(1 mark)

- (ii) The data in the spreadsheet is sorted. State the column on which the sorting feature was applied and the sorting order.

Column A (Shipping Line), sorted in alphabetical/ascending

(2 marks)

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



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- (iii) Write the formula/function to be inserted in Cell H3 to calculate the number of containers shipped during the week for Canva Shipping.

=SUM(B3:G3)

(2 marks)

- (iv) Write the function that would be used to determine the **lowest** number of containers shipped on Wednesday.

=MIN(D3:D7)

(2 marks)

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



0 1 2 2 9 0 2 0 1 7

(c) Figure 1 below is a chart was created using the spreadsheet data of the Logistics Hub.

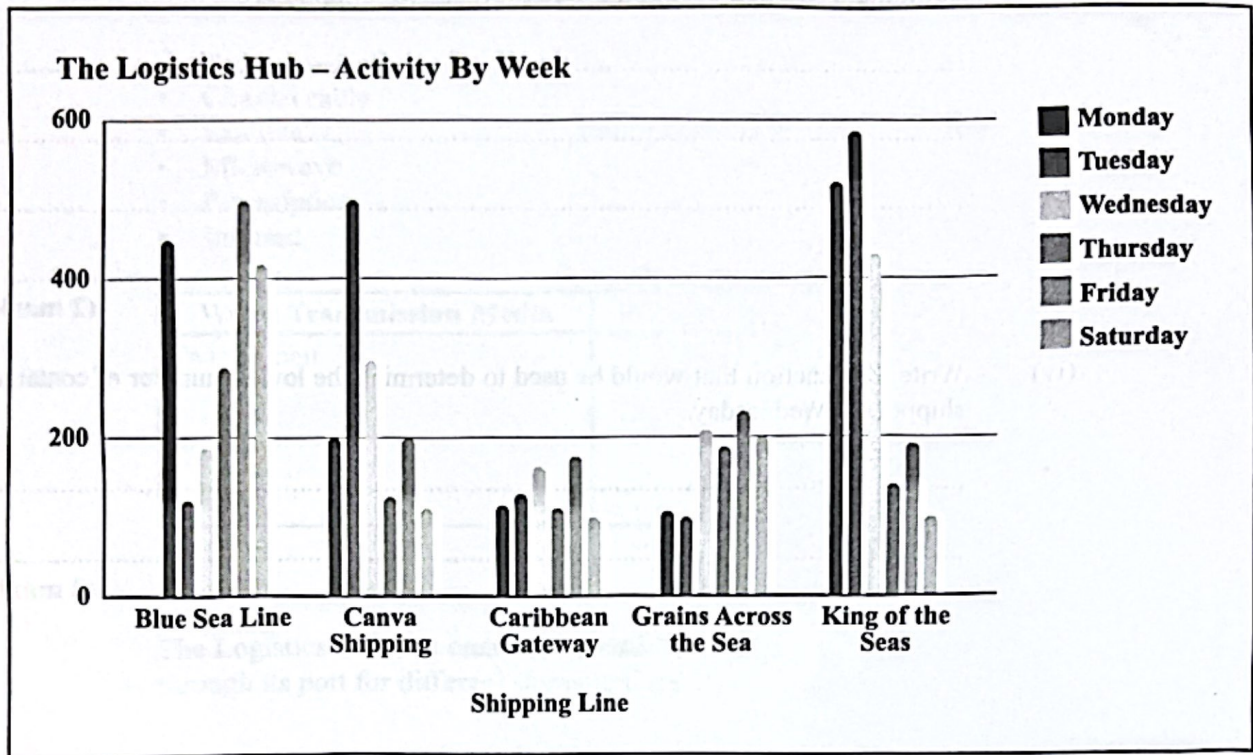


Figure 1. The Logistics Hub – Activity by Week

(i) State the type of chart used to represent the data in Figure 1.

Clustered Column Chart

(1 mark)

(ii) Using the spreadsheet data provided in (b) on page 16, list (in the correct order) the steps required for creating the chart shown in Figure 1.

Highlight the range A2:G7, click the "Insert" TAB, and select "Column"

(2 marks)

GO ON TO THE NEXT PAGE



- (d) Define the term 'syntax error'.

.....
An error in the grammar or rules of a programming language
.....

(1 mark)

- (e) Complete the table below by inserting the data types to be used for EACH of the following variables.

Variable	Data Type
FirstName	String
ItemPrice	Real
Receipt_Number	Integer
Instock	Boolean

(4 marks)

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



(f) Using a **while loop** construct, write pseudocode to produce the following.

- Accept a whole number from the user.
- If the number entered is greater than zero, print "Positive Number".
- If the number entered is less than zero, print "Negative Number".
- If the number entered is zero, print "You have entered a zero. Goodbye".

Declare all variables used.

READ num

WHILE num <> 0 DO

IF num > 0 THEN

PRINT "Positive Number"

ELSE

PRINT "Negative Number"

ENDIF

READ numENDWHILEPRINT "You have entered a zero."

GO ON TO THE NEXT PAGE

01229020/MJ/CSEC 2025



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0 1 2 2 8 0 2 0 2 1